

EE 446

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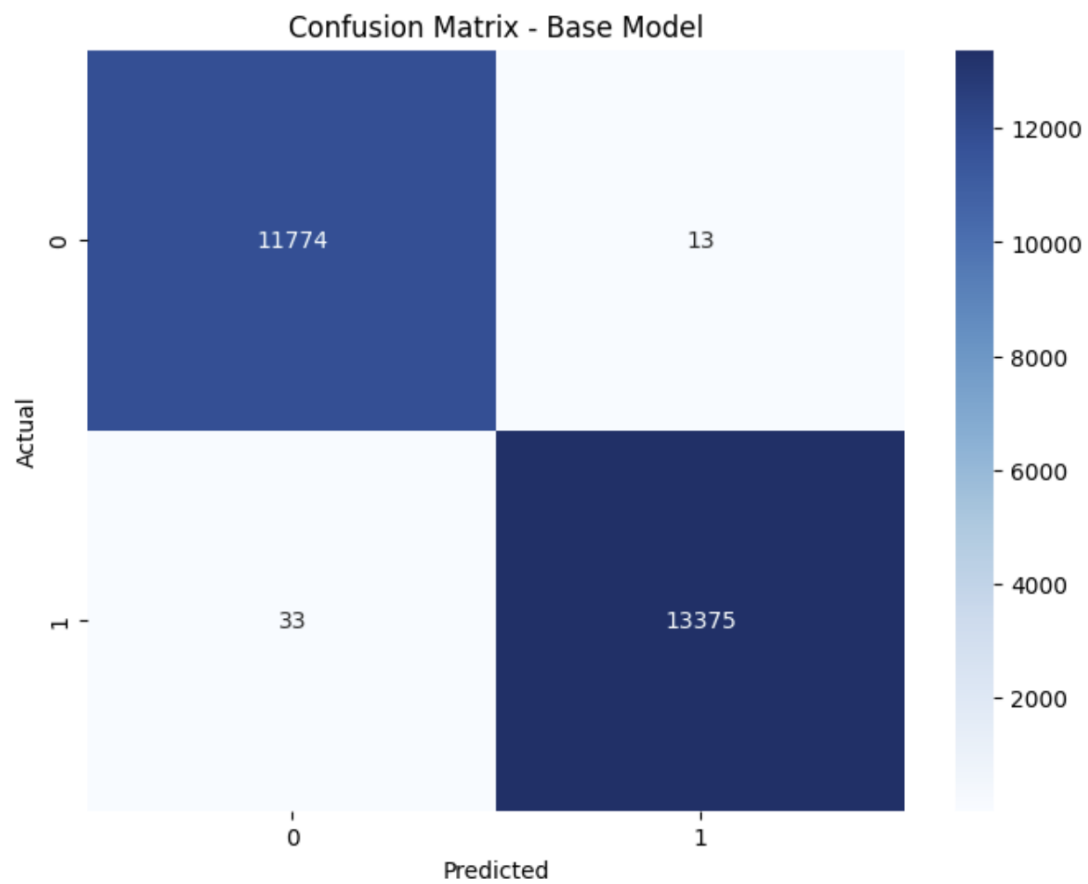
Assignment 2

Problem 3(c)

788/788 [=====] - 1s 1ms/step

Classification Report:

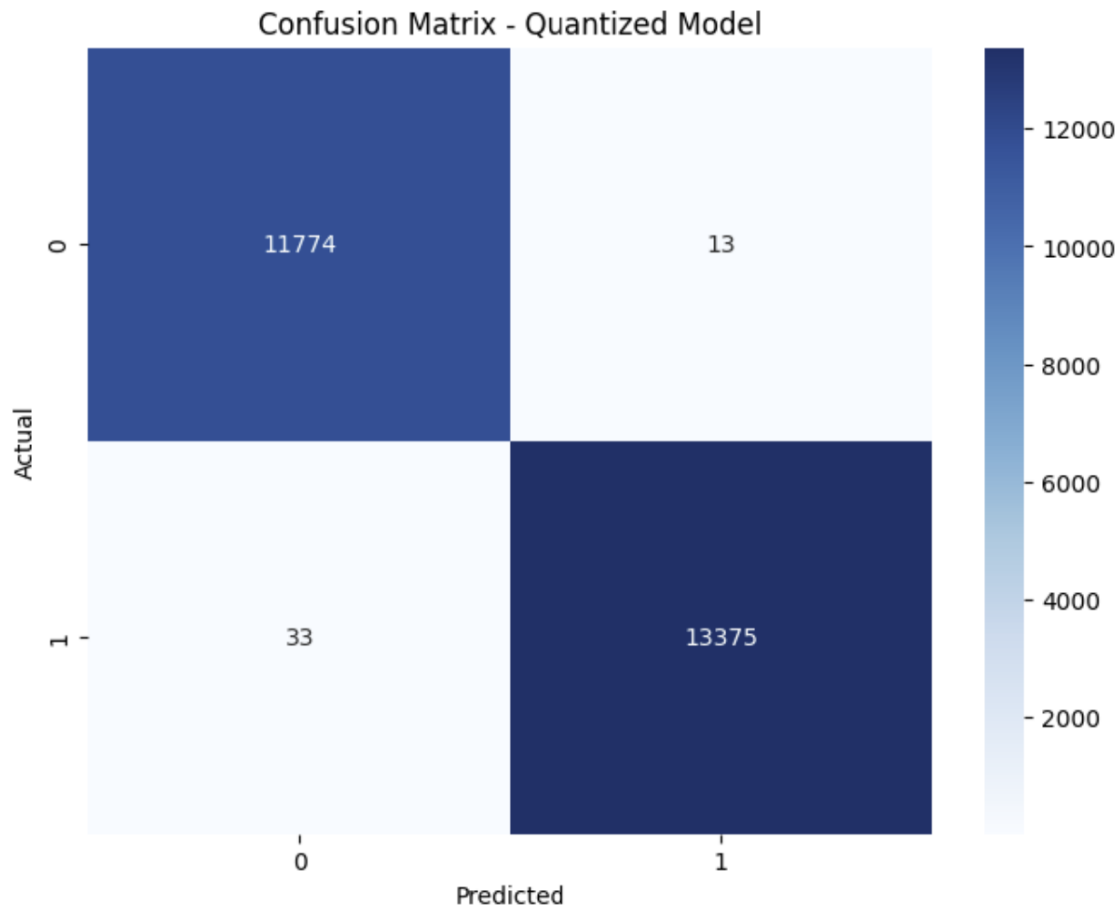
	precision	recall	f1-score	support
0	1.00	1.00	1.00	11787
1	1.00	1.00	1.00	13408
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195



Problem 4(b)

Classification Report (Quantized Model):

	precision	recall	f1-score	support
0	1.00	1.00	1.00	11787
1	1.00	1.00	1.00	13408
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195



Problem 5(c)

```
16:17:06.914 -> Sample #1 | Predicted Class: 1 | Actual Class: 1
16:17:07.897 -> Sample #2 | Predicted Class: 1 | Actual Class: 1
16:17:08.931 -> Sample #3 | Predicted Class: 0 | Actual Class: 0
16:17:09.936 -> Sample #4 | Predicted Class: 0 | Actual Class: 0
16:17:10.944 -> Sample #5 | Predicted Class: 1 | Actual Class: 1
```

Note: For some reason I couldn't get the quantized version of the model to deploy on the nano 33 ble. I think it was because the weights are stored as int8, but activation remains float32 format. Deploying a non-quantized version worked although the model was a little bigger.

Problem 5(d)

```
Sample #1 | Predicted Class: 1 | Actual Class: 1
Sample #2 | Predicted Class: 0 | Actual Class: 0
Sample #3 | Predicted Class: 0 | Actual Class: 0
Sample #4 | Predicted Class: 1 | Actual Class: 1
Sample #5 | Predicted Class: 0 | Actual Class: 0
Sample #6 | Predicted Class: 0 | Actual Class: 0
Sample #7 | Predicted Class: 1 | Actual Class: 1
Sample #8 | Predicted Class: 1 | Actual Class: 1
Sample #9 | Predicted Class: 0 | Actual Class: 0
Sample #10 | Predicted Class: 0 | Actual Class: 0
```